

--Google Play Store Apps Analysis

--1. Market Share Analysis

--Question 1: Which are the top 10 categories by total installs, and what is their percentage share of the total market?

```
WITH CategoryInstalls AS (  
    SELECT  
        Category,  
        SUM(Installs_Numeric) AS Total_Category_Installs  
    FROM v_GooglePlayStore_Cleaned  
    GROUP BY Category  
)  
GlobalTotal AS (  
    SELECT SUM(CAST(Installs_Numeric AS FLOAT)) AS Grand_Total FROM  
v_GooglePlayStore_Cleaned  
)  
SELECT TOP 10  
    c.Category,  
    c.Total_Category_Installs,  
    (c.Total_Category_Installs * 100.0 / g.Grand_Total) AS Market_Share_Percent  
FROM CategoryInstalls c, GlobalTotal g  
ORDER BY c.Total_Category_Installs DESC;
```

	Category	Total_Category_Installs	Market_Share_Percent
1	Education	16894268	11.5576008306345
2	Music & Audio	10764098	7.36386731794659
3	Business	10249581	7.01187916986136
4	Tools	8353038	5.71442804903541
5	Entertainment	8198858	5.60895139292535
6	Lifestyle	7643438	5.22898094061863
7	Books & Reference	7129708	4.87753119004513
8	Food & Drink	5452307	3.72999812197097
9	Personalization	5400693	3.69468827550278
10	Health & Fitness	5321744	3.64067818000899

--Business Insight: Identifies which categories dominate the store and where the highest volume of users is concentrated.

--2. Ad-Support vs. User Satisfaction

--Question 2: Compare the average rating and engagement for apps with ads versus those without, categorized by Free and Paid status.

```
SELECT  
    Is_Free,  
    Is_Ad_Supported,  
    COUNT(*) AS App_Count,  
    AVG(Rating) AS Avg_Rating,  
    AVG(CAST(Installs_Numeric AS FLOAT)) AS Avg_Installs  
FROM v_GooglePlayStore_Cleaned  
GROUP BY Is_Free, Is_Ad_Supported  
ORDER BY Is_Free DESC, Is_Ad_Supported DESC;
```

	Is_Free	Is_Ad_Supported	App_Count	Avg_Rating	Avg_Installs
1	1	0	1172026	0.94740961377987	122.191750012372
2	0	0	27998	1.57963068790628	105.815272519466

--Business Insight: Helps determine if aggressive monetization (Ads) significantly negatively impacts user ratings—crucial for Ad-Tech firms.

--3. The "Elite Three" (Window Functions)

--Question 3: For every category, identify the top 3 apps with the highest number of ratings, provided they maintain a rating of at least 4.0.

```
WITH RankedApps AS (
    SELECT
        App_Name,
        Category,
        Rating,
        Rating_Count,
        Row_Number() OVER (PARTITION BY Category ORDER BY Rating_Count DESC) AS
        Rank_In_Category
    FROM v_GooglePlayStore_Cleaned
    WHERE Rating >= 4.0
)
SELECT TOP 15 * FROM RankedApps
WHERE Rank_In_Category <= 3
ORDER BY Category, Rating Desc;
```

--Business Insight: Action leaders require a perfect 5.0 rating to dominate, whereas high-volume categories like Arcade prioritize total engagement count over absolute rating score.

	App_Name	Category	Rating	Rating_Count	Rank_In_Category
1	Monk Power Archer	Action	5	476	2
2	Kule Savunma - Bird Defense	Action	5	502	1
3	Stickman Hero Gladiator : Epic Battle	Action	4.8	262	3
4	Extreme Jungle Rush	Adventure	4.9	190	3
5	Stuck Ship: Boat Games 2D	Adventure	4.9	210	2
6	Baseball Ball	Adventure	4.9	261	1
7	Wavy Rocket	Arcade	5	311	3
8	CAR DEFENDER - Car game By GCG	Arcade	4.9	502	2
9	Color Spinner: Color Circle Match- Hit & Win Color	Arcade	4.1	904	1
10	Esport T Shirt Design - Gaming T Shirt Maker	Art & Design	5	203	3
11	T Shirt Design Pro - Custom T Shirts	Art & Design	4.9	203	2
12	Mod Bussid Truck Umplung	Art & Design	4.1	7923	1
13	Cars for sale	Auto & Vehicles	5	227	3
14	Oil Reset - Upto 2021	Auto & Vehicles	4.9	318	1
15	Modern Tuk Tuk Auto Rickshaws : Mega Driving Games	Auto & Vehicles	4	233	2

--4. Engagement Intensity Index

--Question 4: Which categories have the most "vocal" users? (Ratio of Rating Count to Total Installs).

```
SELECT top 10
    Category,
    SUM(Rating_Count) AS Total_Ratings,
    SUM(Installs_Numeric) AS Total_Installs,
    (CAST(SUM(Rating_Count) AS FLOAT) / NULLIF(SUM(Installs_Numeric), 0)) * 100 AS
Engagement_Ratio
FROM v_GooglePlayStore_Cleaned
GROUP BY Category
HAVING SUM(Installs_Numeric) > 1000000 -- Focus on significant categories
ORDER BY Engagement_Ratio DESC;
--Business Insight: Photography leads with a 24.5% engagement ratio, nearly 3x
higher than other categories,
--proving that creative and personal tools drive significantly more user feedback
than functional apps like Shopping or Education.
```

--5. "Stale" vs. "Fresh" Apps Analysis

--Question 5: What percentage of apps in each category have not been updated since 2021?

```
SELECT
    Category,
    COUNT(*) AS Total_Apps,
    -- סופרים כמה אפליקציות לא עודכנו מאז 2021
    SUM(CASE
        WHEN [Last_Updated] LIKE '%2010%' OR [Last_Updated] LIKE '%2011%' OR
            [Last_Updated] LIKE '%2012%' OR [Last_Updated] LIKE '%2013%' OR
            [Last_Updated] LIKE '%2014%' OR [Last_Updated] LIKE '%2015%' OR
            [Last_Updated] LIKE '%2016%' OR [Last_Updated] LIKE '%2017%' OR
            [Last_Updated] LIKE '%2018%' OR [Last_Updated] LIKE '%2019%' OR
            [Last_Updated] LIKE '%2020%'
        THEN 1 ELSE 0 END) AS Stale_Apps,
    -- חישוב האחוז ועיגולו ל-2 ספרות אחרי הנקודה
    ROUND((SUM(CASE
        WHEN [Last_Updated] LIKE '%2010%' OR [Last_Updated] LIKE '%2011%' OR
            [Last_Updated] LIKE '%2012%' OR [Last_Updated] LIKE '%2013%' OR
            [Last_Updated] LIKE '%2014%' OR [Last_Updated] LIKE '%2015%' OR
            [Last_Updated] LIKE '%2016%' OR [Last_Updated] LIKE '%2017%' OR
            [Last_Updated] LIKE '%2018%' OR [Last_Updated] LIKE '%2019%' OR
            [Last_Updated] LIKE '%2020%'
        THEN 1.0 ELSE 0 END) / COUNT(*)) * 100, 2) AS Stale_Rate_Percent
FROM [dbo].[GooglePlayStore_Raw]
GROUP BY Category
-- רק קבוצות שיש בהן יותר מ-100 אפליקציות
HAVING COUNT(*) > 100
ORDER BY Stale_Rate_Percent DESC;
```

	Category	Total_Apps	Stale_Apps	Stale_Rate_Percent
1	Comics	2787	1724	61.860000
2	Casino	4978	3049	61.250000
3	Card	8037	4765	59.290000
4	Video Players & Editors	13245	7721	58.290000
5	Simulation	23057	13244	57.440000
6	Photography	34064	19047	55.920000
7	Personalization	87181	48678	55.840000
8	Books & Reference	112511	61596	54.750000
9	Weather	7060	3705	52.480000
10	Role Playing	9932	5138	51.730000
11	Dating	5871	2979	50.740000
12	Board	10400	5241	50.390000
13	Word	8479	4245	50.060000
14	Music	4135	2053	49.650000
15	Parenting	3614	1792	49.580000

--High stagnation in categories like Comics and Casino (60%+) reveals a massive market gap for new, updated apps to disrupt and replace neglected competition.

--6. Value of Editor's Choice

--Question 6: Does being an "Editor's Choice" actually correlate with more installs and better ratings in the Games category?

```
SELECT
    Is_Editors_Choice,
    COUNT(*) AS App_Count,
    AVG(Rating) AS Avg_Rating,
    AVG(CAST(Installs_Numeric AS FLOAT)) AS Avg_Installs
FROM v_GooglePlayStore_Cleaned
WHERE Category IN ('Action', 'Arcade', 'Puzzle', 'Simulation', 'Role Playing',
'Card', 'Casino', 'Board', 'Word', 'Racing', 'Strategy')
GROUP BY Is_Editors_Choice;
```

	Is_Editors_Choice	App_Count	Avg_Rating	Avg_Installs
1	0	26052	1.57541455550438	95.9705204974666
2	1	68770	1.60888905045804	120.403606223644

--Business Insight: Editor's Choice apps demonstrate a clear competitive edge, securing higher user satisfaction and a 25% increase in average installs compared to non-featured apps.

--7. High-Performing Developers

--Question 7: Identify developers with more than 5 apps and rank them by their weighted average rating (weighted by installs).

```
SELECT
    Developer_Id,
    COUNT(App_Name) AS Total_Apps,
    SUM(Rating * Minimum_Installs) / NULLIF(SUM(Minimum_Installs), 0) AS
Weighted_Rating,
    SUM(Minimum_Installs) AS Total_Installs
FROM v_GooglePlayStore_Cleaned
GROUP BY Developer_Id
HAVING COUNT(App_Name) > 5
```

ORDER BY Weighted_Rating DESC;

	Developer_Id	Total_Apps	Weighted_Rating	Total_Installs
1	Sebastian V.	6	5	280
2	Astro Game Studio	6	4.988888888888889	900
3	Online FM AM Radio Stations	7	4.97894736842105	1900
4	sincApp Studio	6	4.97237569060774	181
5	Quizinator	8	4.965	200
6	CHABOD MULTIMEDIA	7	4.96360759493671	632
7	StepUpApps	7	4.96323529411765	1360
8	Infanion BVBA	7	4.95714285714286	700
9	InfyOm Technologies	6	4.95714285714286	1400
10	SMTech Apps	6	4.95419847328244	1310
11	Arnau Beltran Mateu	8	4.94791666666667	2112
12	CCS ELETRONICOS	9	4.93655913978495	930
13	Islamic Studio	7	4.92105263157895	1900
14	FL Design	6	4.92	1000

	Developer_Id	Total_Apps	Weighted_Rating	Total_Installs
5009	Ostad Gugel	6	2.69230769230769	910
5010	Sapphire Software Solutions	41	2.69225302061123	7035
5011	mydigitalearth.com	36	2.69122477709995	4262
5012	Technogleam	11	2.69117647058824	680
5013	fabklean	6	2.69090909090909	550
5014	Tak Team Studio	18	2.69086021505376	3720
5015	Mitra Sukses	8	2.69072164948454	970
5016	SS PRO	20	2.69041769041769	4070
5017	Ilog Tecnologia	6	2.69005847953216	1710
5018	Pixaflip Technologies	7	2.68987341772152	316
5019	Lohit Kumar	7	2.68980477223427	461
5020	aAa Designer	6	2.68888888888889	450
5021	Classera Inc.	105	2.68883300735774	10737
5022	AJV	8	2.68867924528302	1060

--Business Insight: Content over Volume: Portfolio size is not a predictor of success. High ratings are driven by the intrinsic quality of the application itself,

--as both small and large developers can experience poor performance.

--8. App Size vs. Rating Performance

--Question 8: Does the size of the app impact user ratings? (Segmented into Small, Medium, Large, Huge).

```
WITH SizeSegmentation AS (
    SELECT
        Rating,
        CASE
            WHEN Size LIKE '%k' THEN 'Small (<1MB)'
            WHEN TRY_CAST(REPLACE(Size, 'M', '' ) AS FLOAT) < 20 THEN 'Medium (1-20MB)'
            WHEN TRY_CAST(REPLACE(Size, 'M', '' ) AS FLOAT) < 100 THEN 'Large (20-100MB)'
```

```

        ELSE 'Huge (>100MB)'
    END AS Size_Bucket
FROM v_GooglePlayStore_Cleaned
WHERE Size NOT IN ('Varies with device', 'N/A')
)
SELECT
    Size_Bucket,
    AVG(Rating) AS Avg_Rating,
    COUNT(*) AS App_Count
FROM SizeSegmentation
GROUP BY Size_Bucket
ORDER BY Avg_Rating DESC;

```

	Size_Bucket	Avg_Rating	App_Count
1	Large (20-100MB)	1.05482895077265	340136
2	Small (<1MB)	0.973677972797562	17719
3	Medium (1-20MB)	0.918646367739025	809895
4	Huge (>100MB)	0.799241693978536	15561

--Business Insight: Size-to-Satisfaction Threshold: Ratings peak for 'Large' apps (20-100MB) but drop sharply for 'Huge' apps (>100MB), indicating a performance penalty for excessive file sizes.

--9. In-App Purchase (IAP) Monetization Sentiment

--Question 9: What is the rating difference between apps that offer In-App Purchases and those that don't, across the top 10 categories?

```

SELECT Top 10
    Category,
    AVG(CASE WHEN Has_In_App_Purchases = 1 THEN Rating ELSE NULL END) AS
Avg_Rating_With_IAP,
    AVG(CASE WHEN Has_In_App_Purchases = 0 THEN Rating ELSE NULL END) AS
Avg_Rating_No_IAP
FROM v_GooglePlayStore_Cleaned
GROUP BY Category
ORDER BY (AVG(CASE WHEN Has_In_App_Purchases = 1 THEN Rating ELSE NULL END)) DESC;

```

	Category	Avg_Rating_With_IAP	Avg_Rating_No_IAP
1	Music	5	0.964179104477612
2	Racing	2.45	1.52830903790087
3	Role Playing	2.17142857142857	1.77199040767386
4	Photography	2.11818181818182	0.720572938165607
5	Action	2.09047619047619	1.54670224297507
6	Trivia	2.02258064516129	1.24765722430073
7	Events	2.01578947368421	0.70904757074754
8	Educational	1.93529411764706	1.11926470588235
9	Sports	1.87765957446809	0.982352941176472
10	Arcade	1.78409090909091	1.80961538461538

--Business Insight: Monetization as a Quality Signal: Apps with IAP consistently outscore free-only models, showing that users value the superior content and production quality typical of paid apps.

```
--10. Audience Demographics by Installs
--Question 10: Within the top 3 most installed categories, how are installs
distributed among Content Ratings (Everyone, Teen, Mature)?
WITH TopCategories AS (
    SELECT TOP 3 Category
    FROM v_GooglePlayStore_Cleaned
    GROUP BY Category
    ORDER BY SUM(Installs_Numeric) DESC
)
SELECT
    Category,
    Content_Rating,
    SUM(Installs_Numeric) AS Total_Installs,
    SUM(SUM(Installs_Numeric)) OVER (PARTITION BY Category) AS Total_Cat_Installs
FROM v_GooglePlayStore_Cleaned
WHERE Category IN (SELECT Category FROM TopCategories)
AND Content_Rating IN ('Everyone', 'Teen', 'Mature 17+', 'Everyone 10+', 'Adults
only 18+', 'Unrated')
GROUP BY Category, Content_Rating
ORDER BY Category, Total_Installs DESC;
```

	Category	Content_Rating	Total_Installs	Total_Cat_Installs
1	Business	Everyone	409984	420880
2	Business	Teen	7956	420880
3	Business	Mature 17+	2840	420880
4	Business	Everyone 10+	100	420880
5	Education	Everyone	236321	240032
6	Education	Everyone 10+	1810	240032
7	Education	Teen	1141	240032
8	Education	Mature 17+	760	240032
9	Music & Audio	Everyone	59867	73392
10	Music & Audio	Teen	10015	73392
11	Music & Audio	Mature 17+	3500	73392
12	Music & Audio	Everyone 10+	10	73392

--Business Insight: The 'Everyone' rating overwhelmingly dominates installs across all top categories, proving that broad accessibility is the most effective strategy for maximizing market reach.